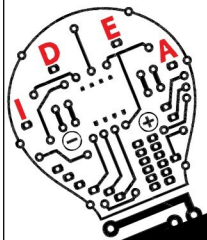


Original, innovative circuit ideas are invited

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also remembers charges.

A flock of small, single-use drones capable of making precise deliveries or completing other military missions and then vapourising into thin air sounds suspiciously like science fiction, but such devices have already been developed and are being brought into the realm of reality. One intriguing example of these ephemeral materials is small polymer panels that sublimate directly from solid to gas.

So far, the disappearing antenna is the most impressive application of this technology. One minute, it is an antenna broadcasting important coordinates or whatever. Drop in a solution, though, and the next minute it is gone—nothing but a few flecks of metal remain.

Another device is a blue LED. It is bright and shiny, and then it is gone, almost without any trace.

Researchers hope to develop more sophisticated devices like a credit card that could dissolve when lost by just sending a signal from a smartphone to start the process.

A new technology can make computer chips self-destruct when remotely triggered. The new method uses silicon computer wafers attached to a piece of tempered glass that shatters into smithereens when heated in one spot. Heat can be turned on via a remote, which, in the future, could conceivably be triggered by anything using Wi-Fi to a radio frequency signal.

Methodology

Researchers have created electronics that will self-destruct on command, and the technology uses radio frequency, acid and a layer of wax on the circuit to let the devices melt with an application of heat or on receiving a signal from a remote device.

They have dissolved electronics in water, which could be used for biomedical implants using heat as the trigger. Heat-triggered devices use magnesium circuits printed on

very thin, flexible materials. They have embedded a weak acid in a bit of wax on the circuit. When wax is heated, acid is released and dissolves the components. To remotely trigger the reaction, researchers have installed a heating coil that the radio signal turns on. That, in turn, melts the wax.

Researchers can control how fast the device degrades by tuning the thickness of wax, concentration of acid and temperature. They can now design a device to self-destruct within 20 seconds to a couple of minutes after heat is applied. Devices can also degrade in steps by encasing different parts in waxes with different melting temperatures.

This gives precise control over parts of a device that are operative, creating possibilities for sophisticated devices that can sense something in the environment and respond to it. To remotely trigger the reaction, researchers have embedded a radio frequency receiver and an inductive heating coil in the device. You can send a signal to cause the coil to heat up, which melts the wax and dissolves the device.

Conclusion

Transient electronics or self-destructing electronic systems work until these are no longer needed, at which point these dissolve completely—dissolution is triggered by even ordinary water in their operating environment. The most immediate application envisioned by developers is for medical devices that dissolve in the human body in a set period of time after being implanted.

Researchers have pulled off a disappearing act for electronic devices that could change the way we think of our gadgets, as well as introduce new capabilities for medical implants and spy gear. In the future, they envision more complex devices that could be adjustable in real time or responsive to changes in their environment such as chemistry, light or pressure. **EFY**